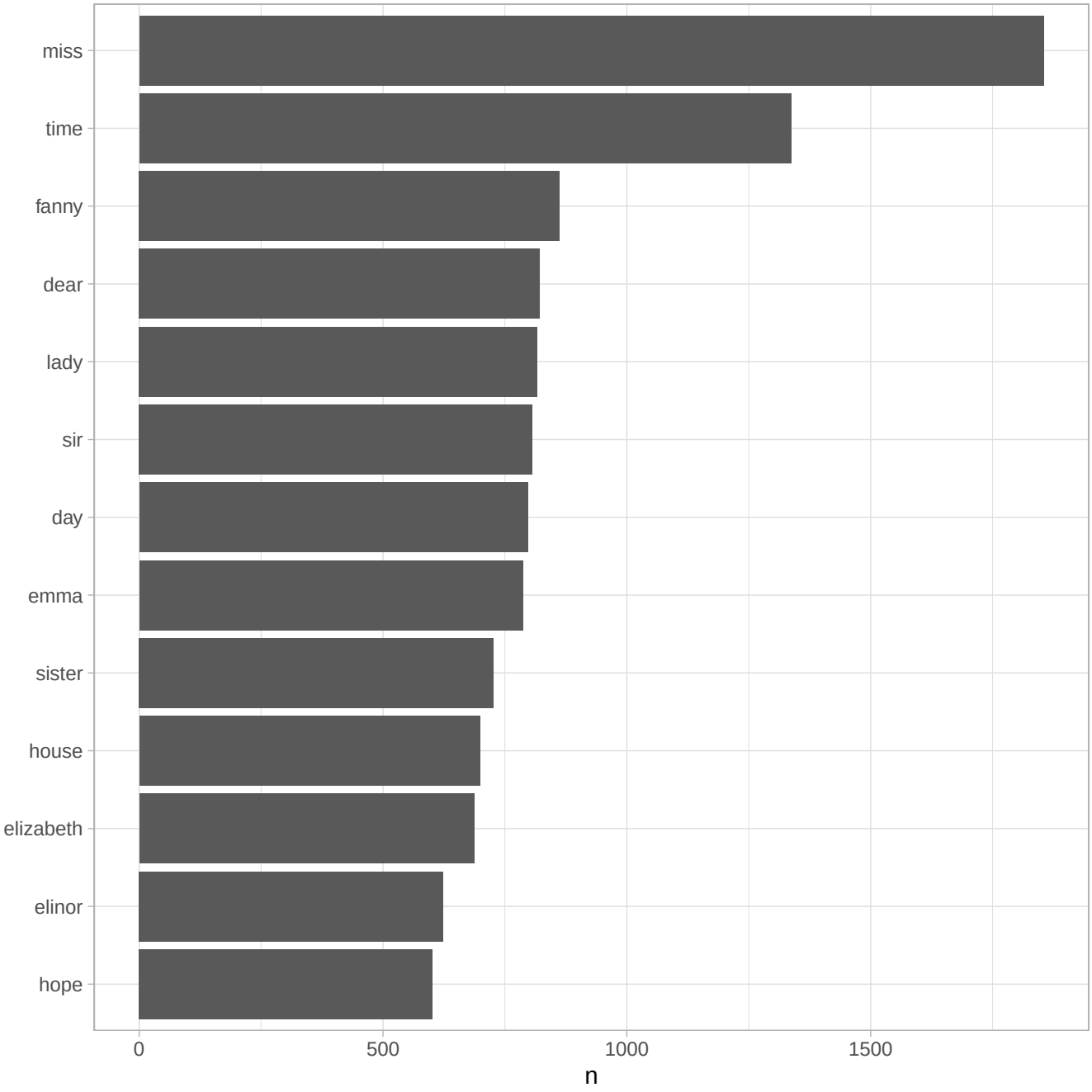
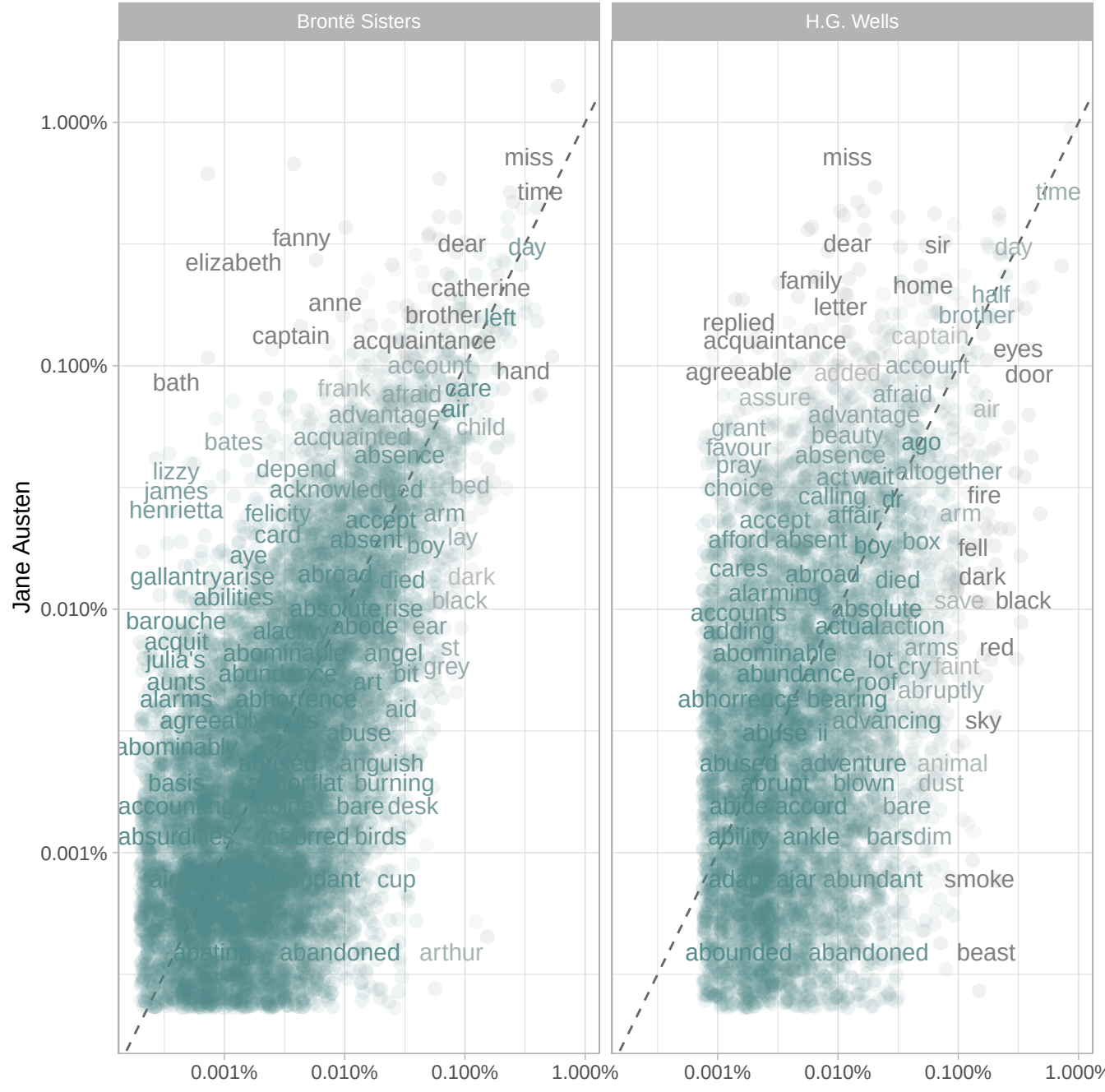
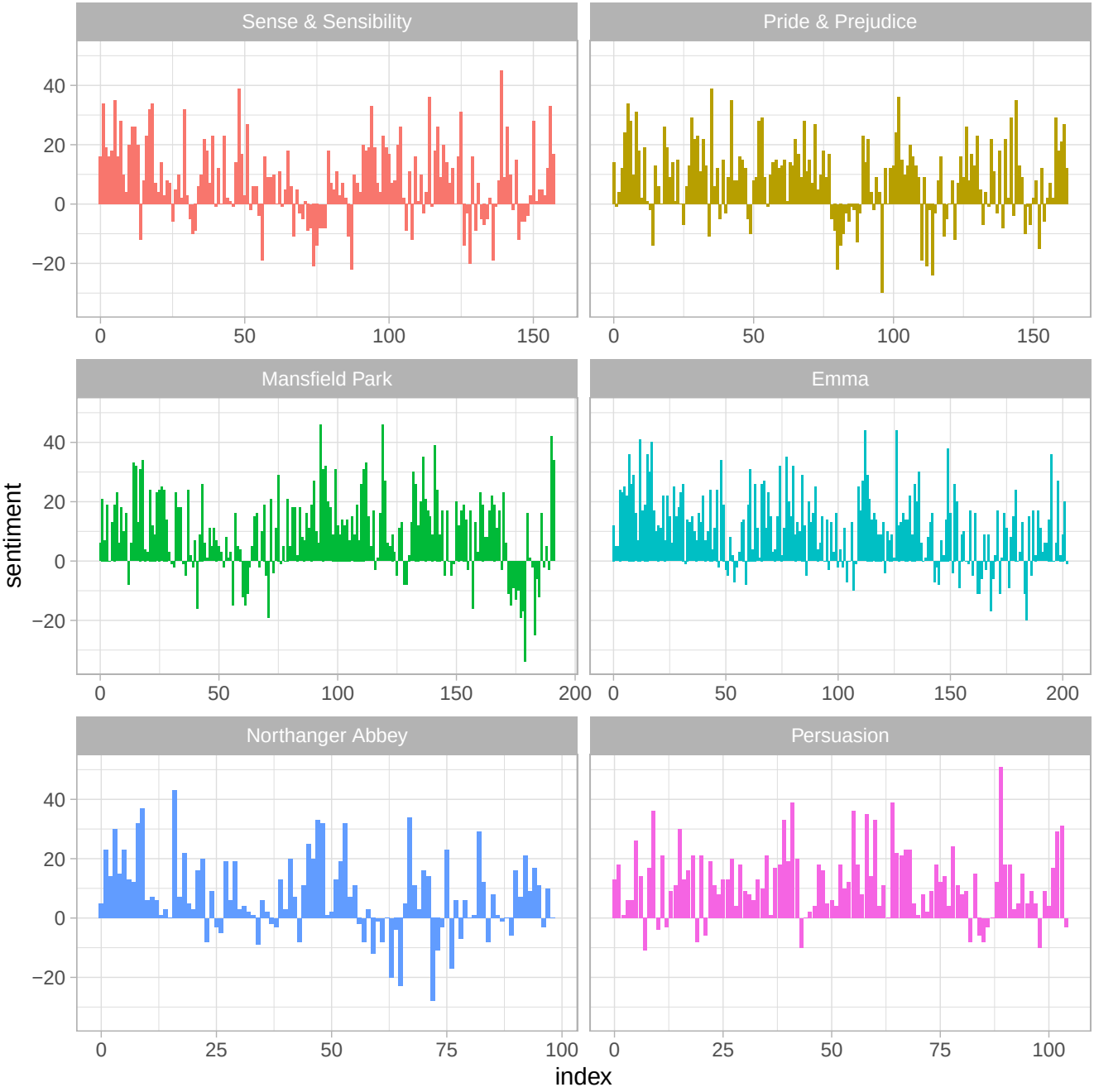
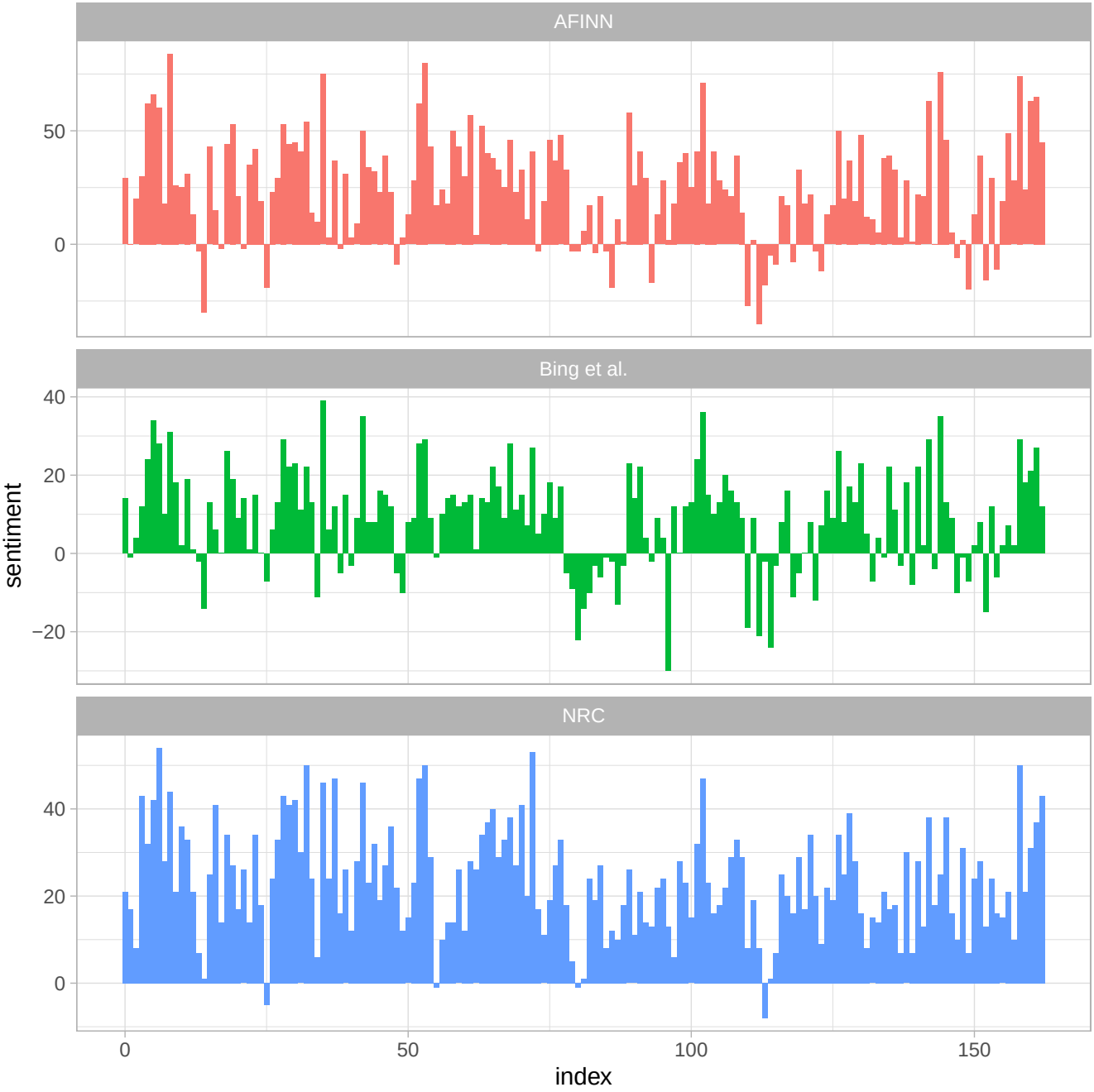
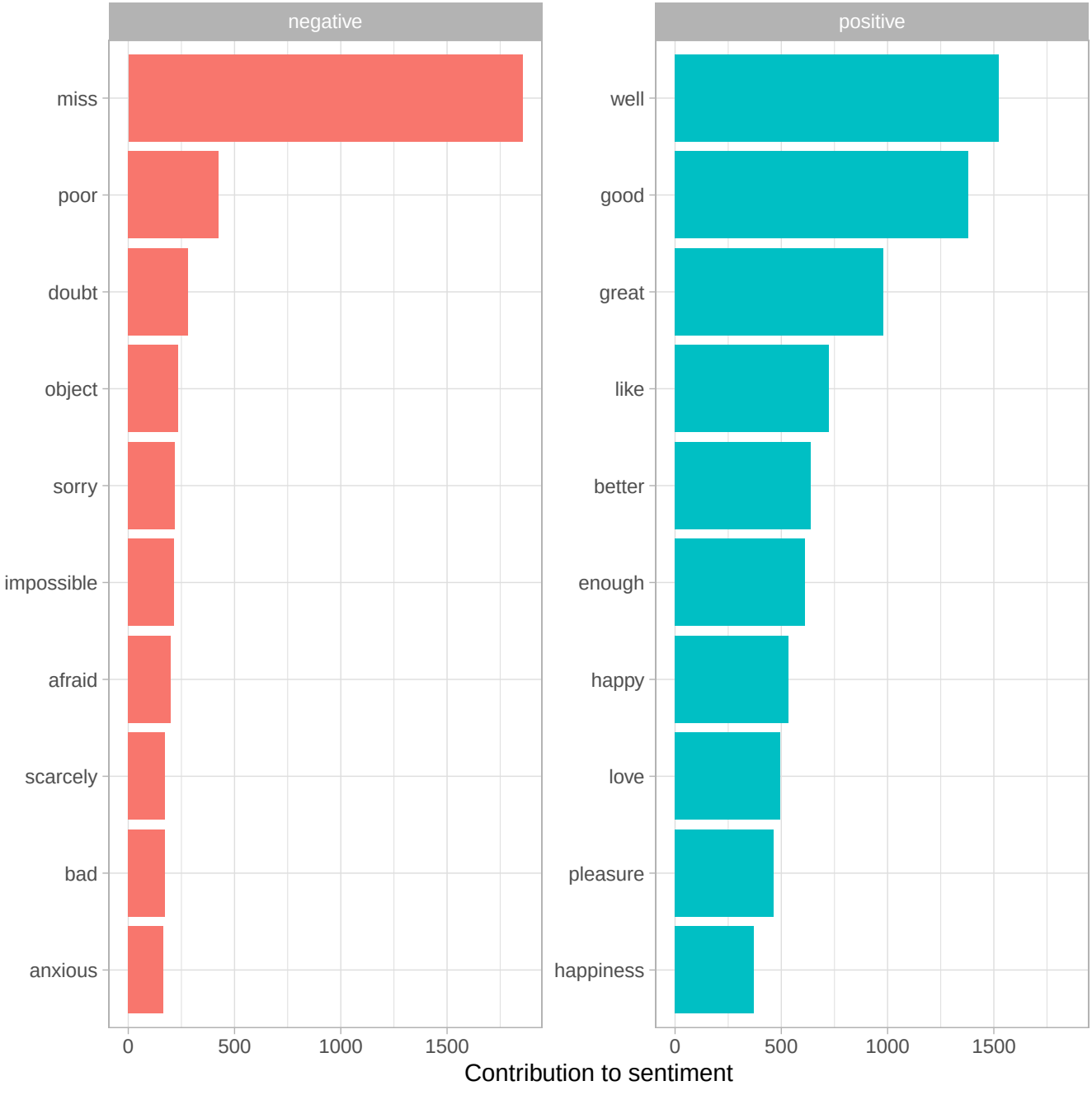
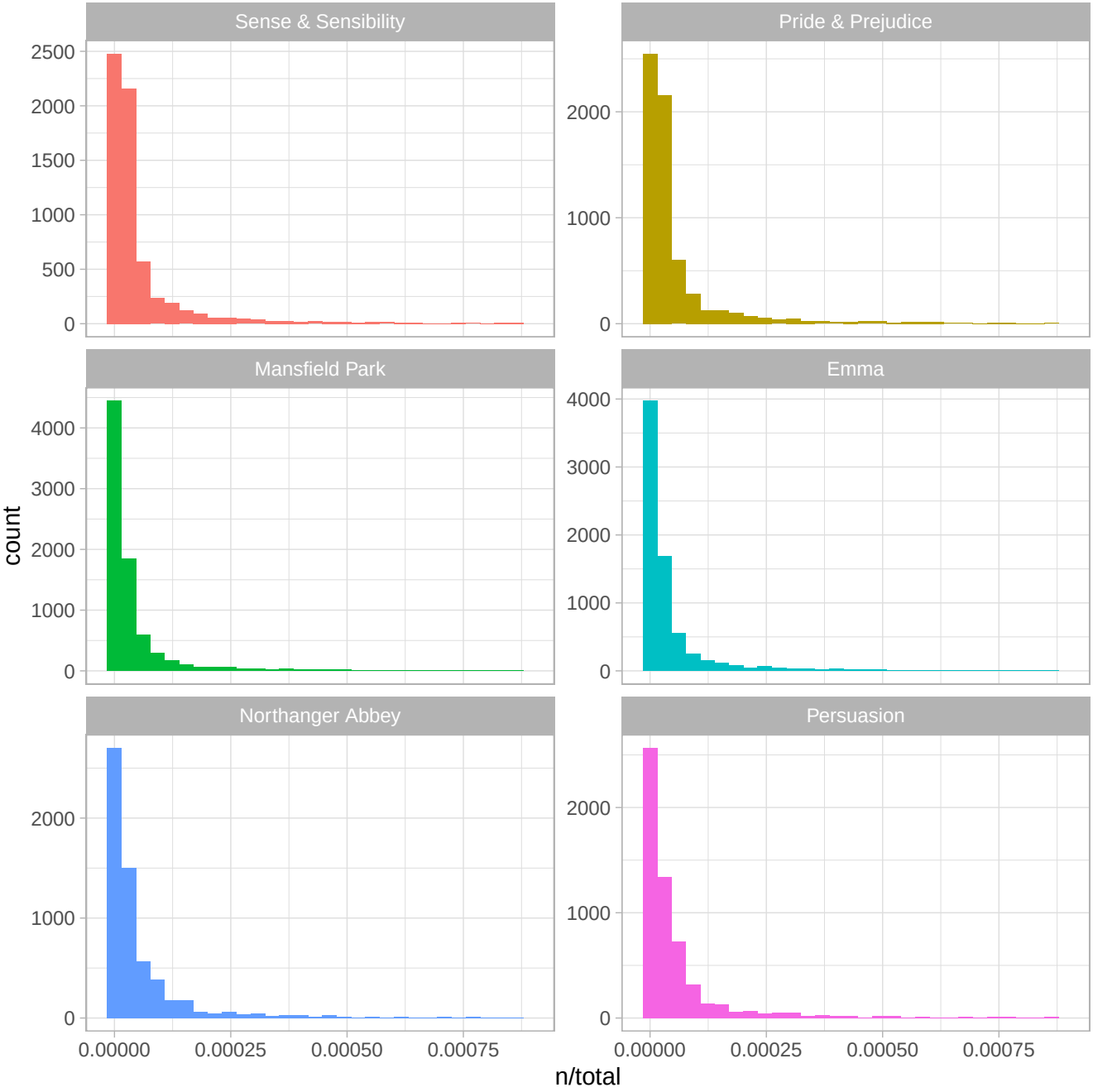

miss
time
fanny
dear
lady
sir
day
emma
sister
house
elizabeth
elinor
hope
0
500
1000
1500
n

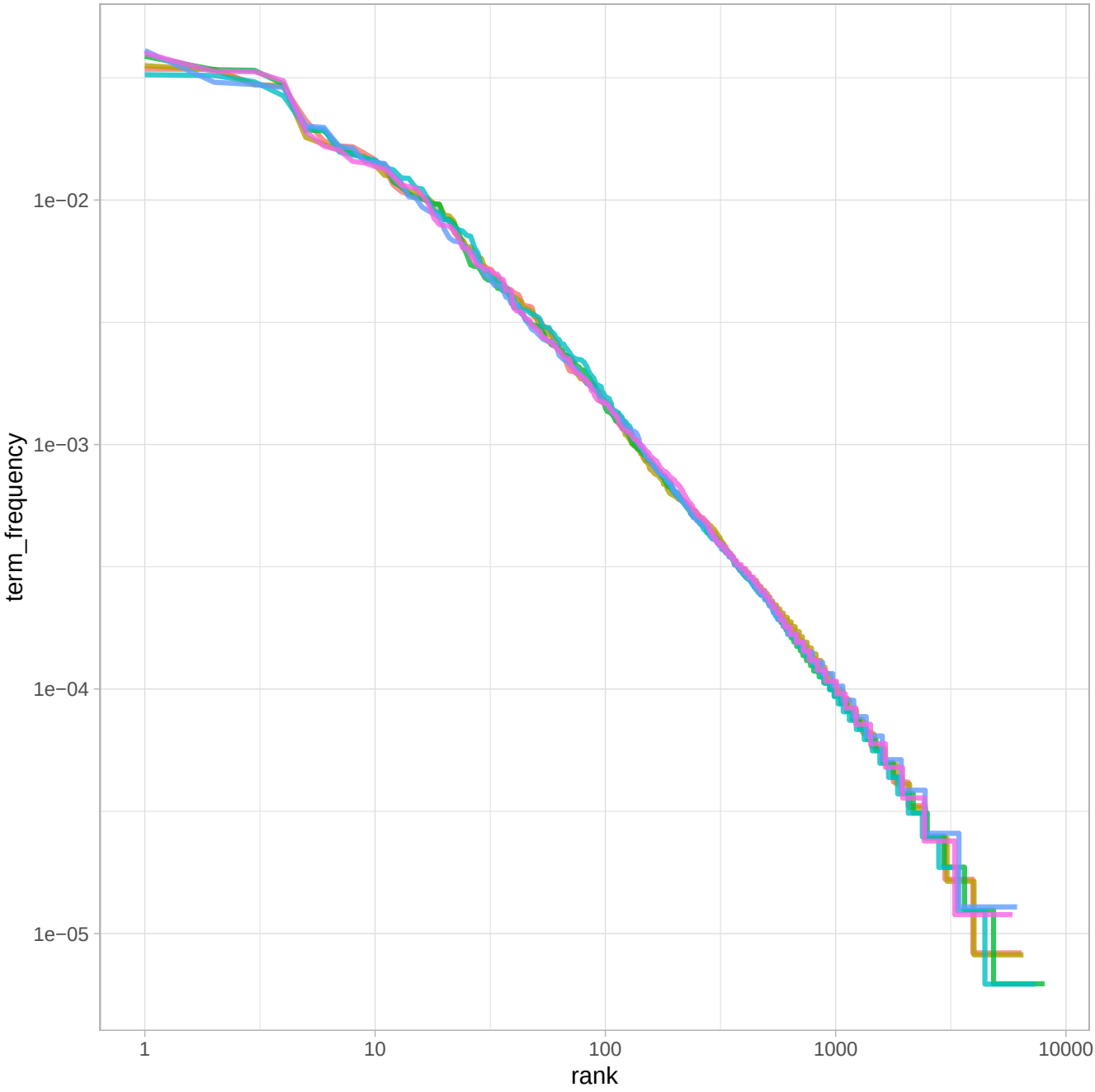

Brontë Sisters
H.G. Wells
Jane Austen
1.000%
0.100%
0.010%
0.001%
miss
time
fanny
dear
day
elizabeth
catherine
anne
brother
left
captain
acquaintance
account
hand
bath
frank
afraid
care
advantage
air
child
bates
acquainted
absence
lizzy
depend
acknowledged
bed
james
henrietta
felicity
accept
arm
card
absent
boy
lay
aye
gallantry
arise
abroad
died
dark
abilities
absolute
rise
black
barouche
abode
ear
acquit
julia's
abominable
angel
st
aunts
abundance
art
bit
grey
alarms
abhorrence
aid
agreeable
abuse
abominably
anguish
basis
flat
burning
accounting
bare
desk
absurdities
abhorred
birds
abundant
cup
abating
abandoned
arthur
sir
family
home
half
letter
replied
acquaintance
captain
eyes
agreeable
added
account
door
assure
afraid
advantage
air
grant
beauty
ago
favour
absence
pray
act
wait
altogether
choice
calling
dr
fire
accept
affair
arm
afford
absent
boy
box
fell
cares
abroad
died
dark
alarming
save
black
accounts
absolute
adding
actual
action
abominable
arms
red
lot
cry
faint
abundance
roof
abruptly
abhorrence
bearing
advancing
sky
abuse
ii
abused
adventure
animal
abrupt
blown
dust
abide
accord
bare
ability
ankle
barsdim
adapt
ajar
abundant
smoke
abounded
abandoned
beast
0.001%
0.010%
0.100%
1.000%

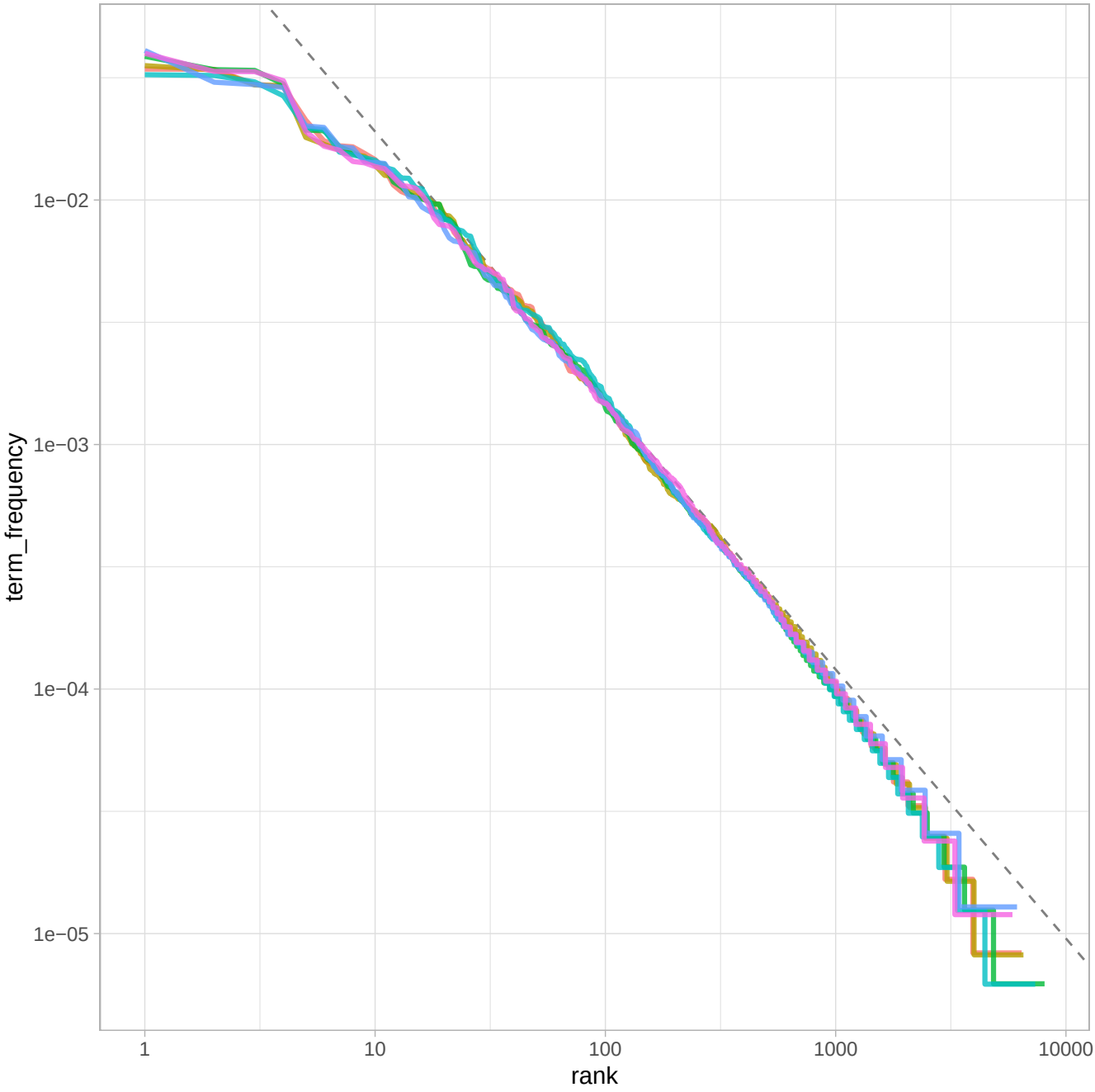

Sense & Sensibility
Pride & Prejudice
Mansfield Park
Emma
Northanger Abbey
Persuasion
sentiment
index
40
20
0
−20
0
50
100
150
200
25
75

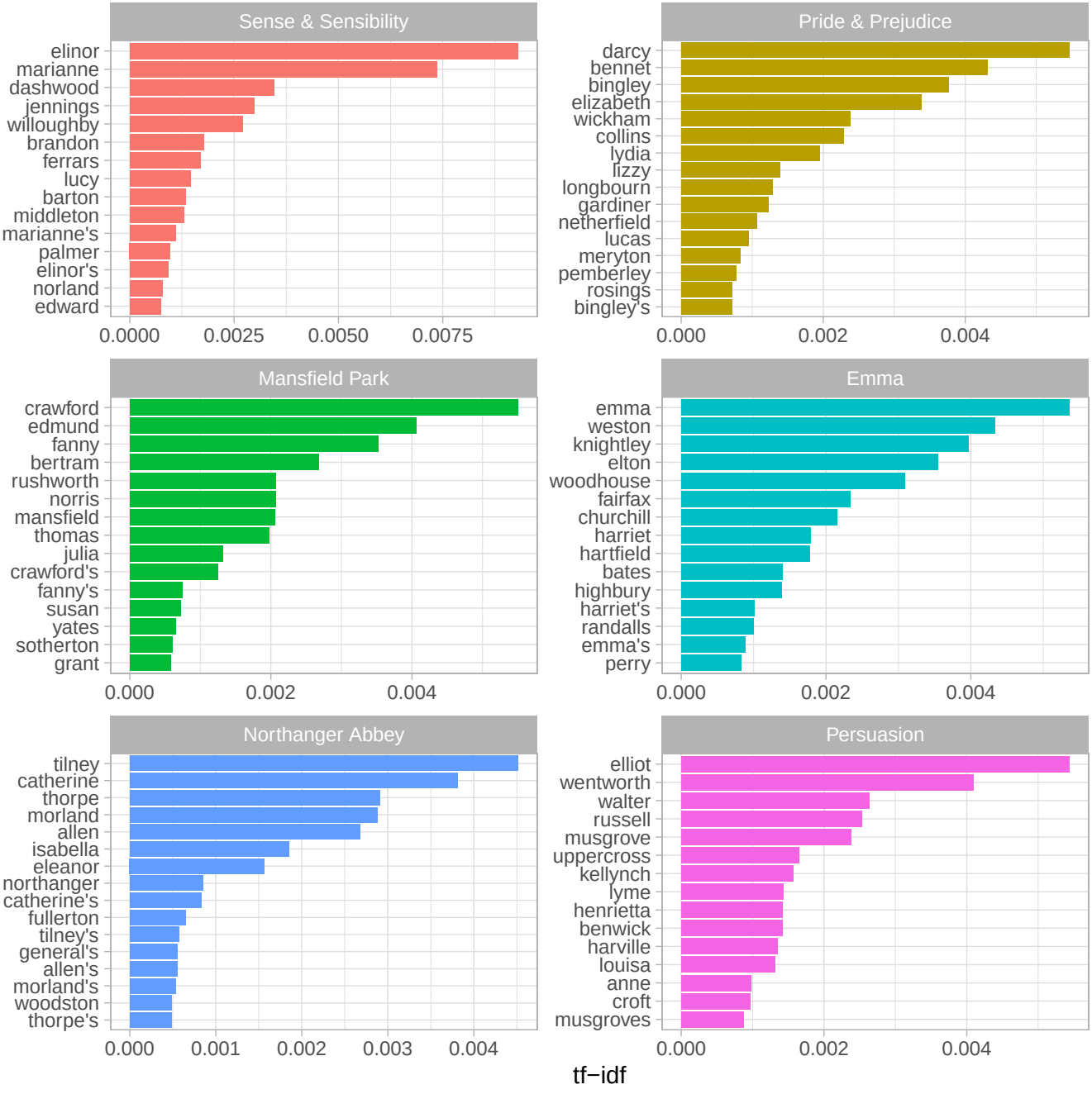

AFINN
Bing et al.
NRC
sentiment
index
50
0
40
20
0
−20
40
20
0
0
50
100
150

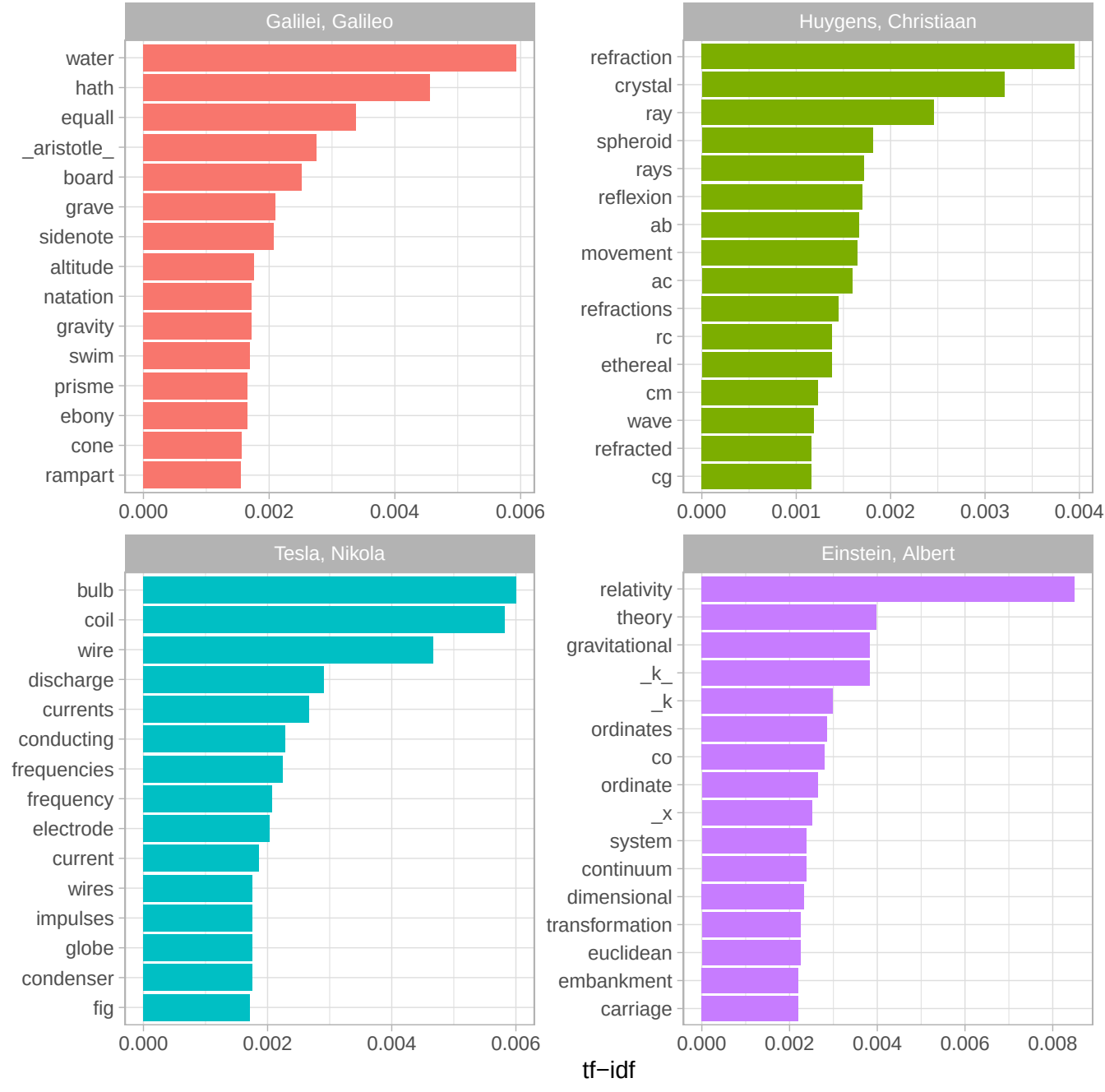

negative
positive
miss
poor
doubt
object
sorry
impossible
afraid
scarcely
bad
anxious
well
good
great
like
better
enough
happy
love
pleasure
happiness
0
500
1000
1500
Contribution to sentiment

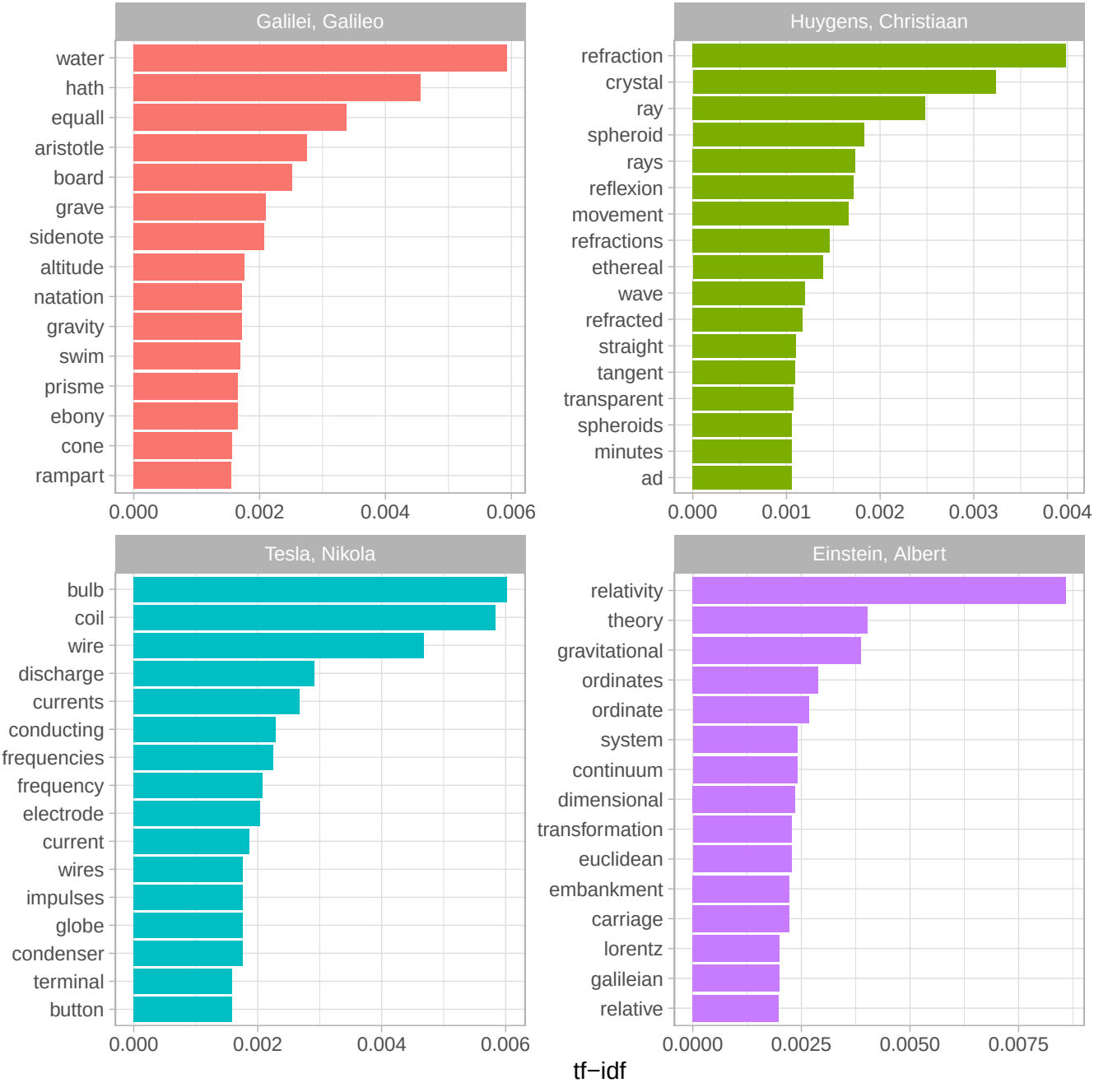

Sense & Sensibility
Pride & Prejudice
Mansfield Park
Emma
Northanger Abbey
Persuasion
count
n/total
0
500
1000
1500
2000
2500
3000
4000
0.00000
0.00025
0.00050
0.00075

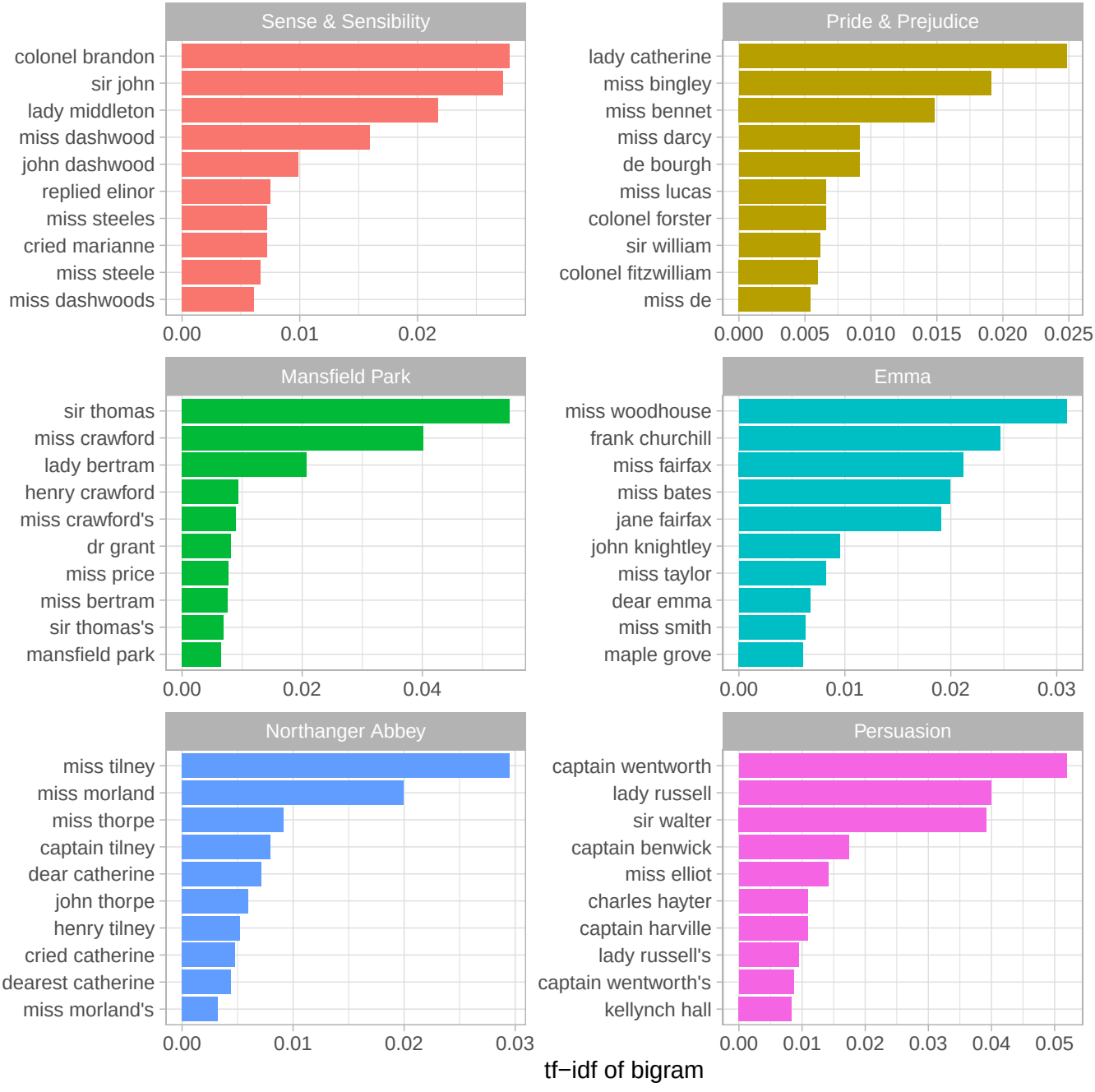

term_frequency
1e−02
1e−03
1e−04
1e−05
1
10
100
1000
10000
rank

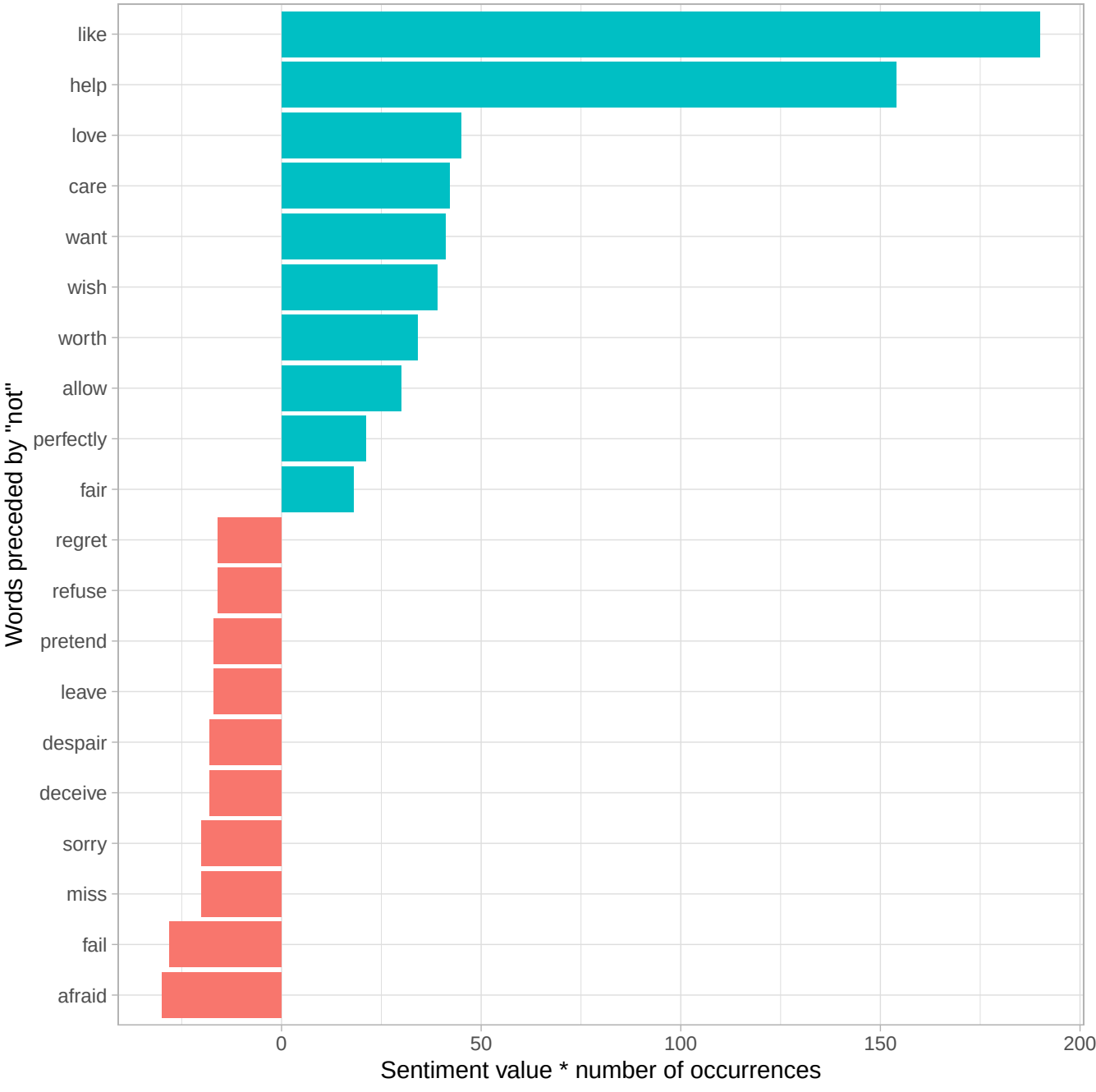

term_frequency
1e−02
1e−03
1e−04
1e−05
1
10
100
1000
10000
rank

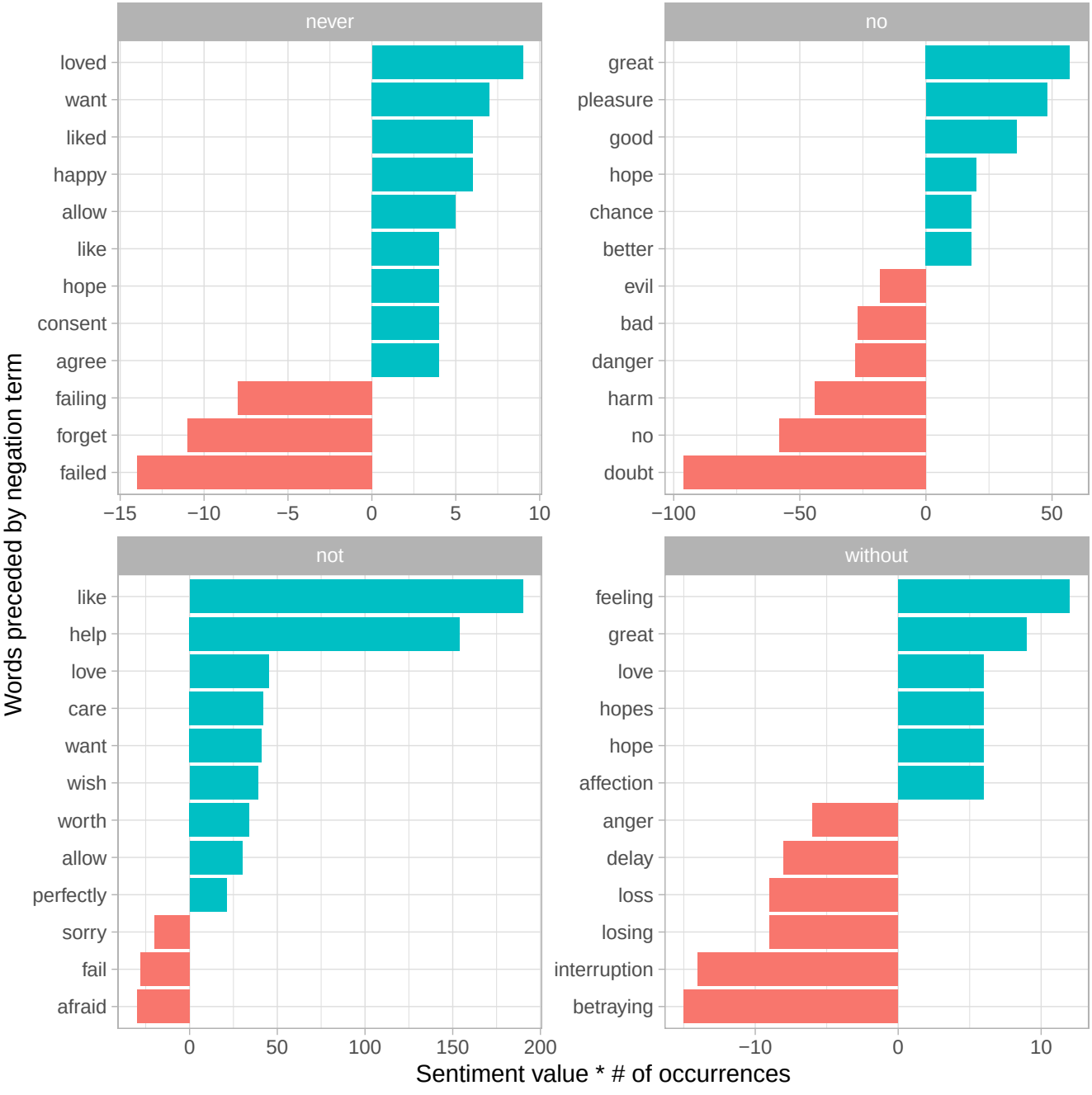

Sense & Sensibility
elinor
marianne
dashwood
jennings
willoughby
brandon
ferrars
lucy
barton
middleton
marianne's
palmer
elinor's
norland
edward
0.0000
0.0025
0.0050
0.0075
Pride & Prejudice
darcy
bennet
bingley
elizabeth
wickham
collins
lydia
lizzy
longbourn
gardiner
netherfield
lucas
meryton
pemberley
rosings
bingley's
0.000
0.002
0.004
Mansfield Park
crawford
edmund
fanny
bertram
rushworth
norris
mansfield
thomas
julia
crawford's
fanny's
susan
yates
sotherton
grant
0.000
0.002
0.004
Emma
emma
weston
knightley
elton
woodhouse
fairfax
churchill
harriet
hartfield
bates
highbury
harriet's
randalls
emma's
perry
0.000
0.002
0.004
Northanger Abbey
tilney
catherine
thorpe
morland
allen
isabella
eleanor
northanger
catherine's
fullerton
tilney's
general's
allen's
morland's
woodston
thorpe's
0.000
0.001
0.002
0.003
0.004
Persuasion
elliot
wentworth
walter
russell
musgrove
uppercross
kellynch
lyme
henrietta
benwick
harville
louisa
anne
croft
musgroves
0.000
0.002
0.004
tf−idf

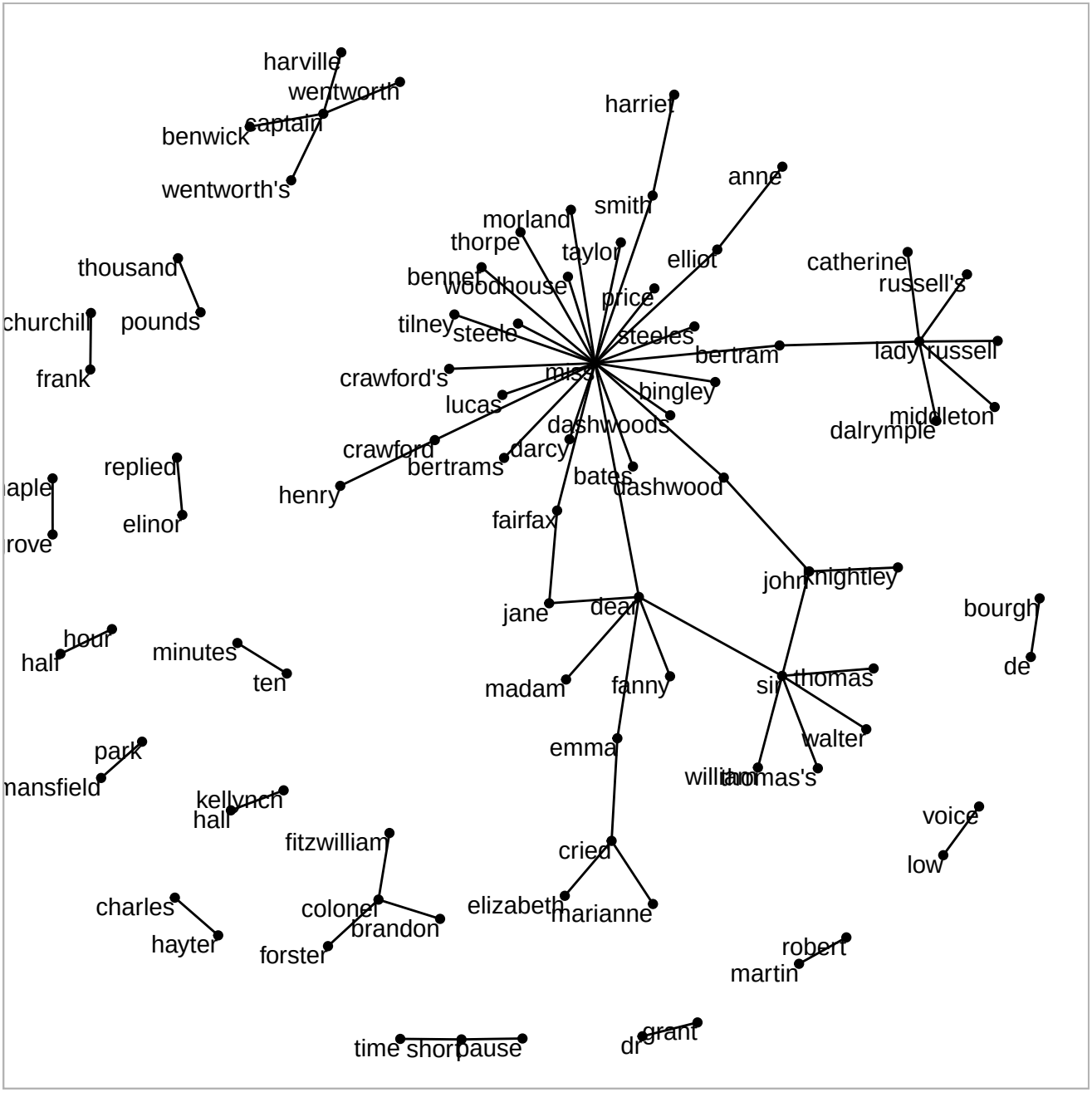

Galilei, Galileo
water
hath
equall
aristotle
board
grave
sidenote
altitude
natation
gravity
swim
prisme
ebony
cone
rampart
0.000
0.002
0.004
0.006
Huygens, Christiaan
refraction
crystal
ray
spheroid
rays
reflexion
ab
movement
ac
refractions
rc
ethereal
cm
wave
refracted
cg
0.000
0.001
0.002
0.003
0.004
Tesla, Nikola
bulb
coil
wire
discharge
currents
conducting
frequencies
frequency
electrode
current
wires
impulses
globe
condenser
fig
0.000
0.002
0.004
0.006
Einstein, Albert
relativity
theory
gravitational
_k_
_k
ordinates
co
ordinate
_x
system
continuum
dimensional
transformation
euclidean
embankment
carriage
0.000
0.002
0.004
0.006
0.008
tf−idf

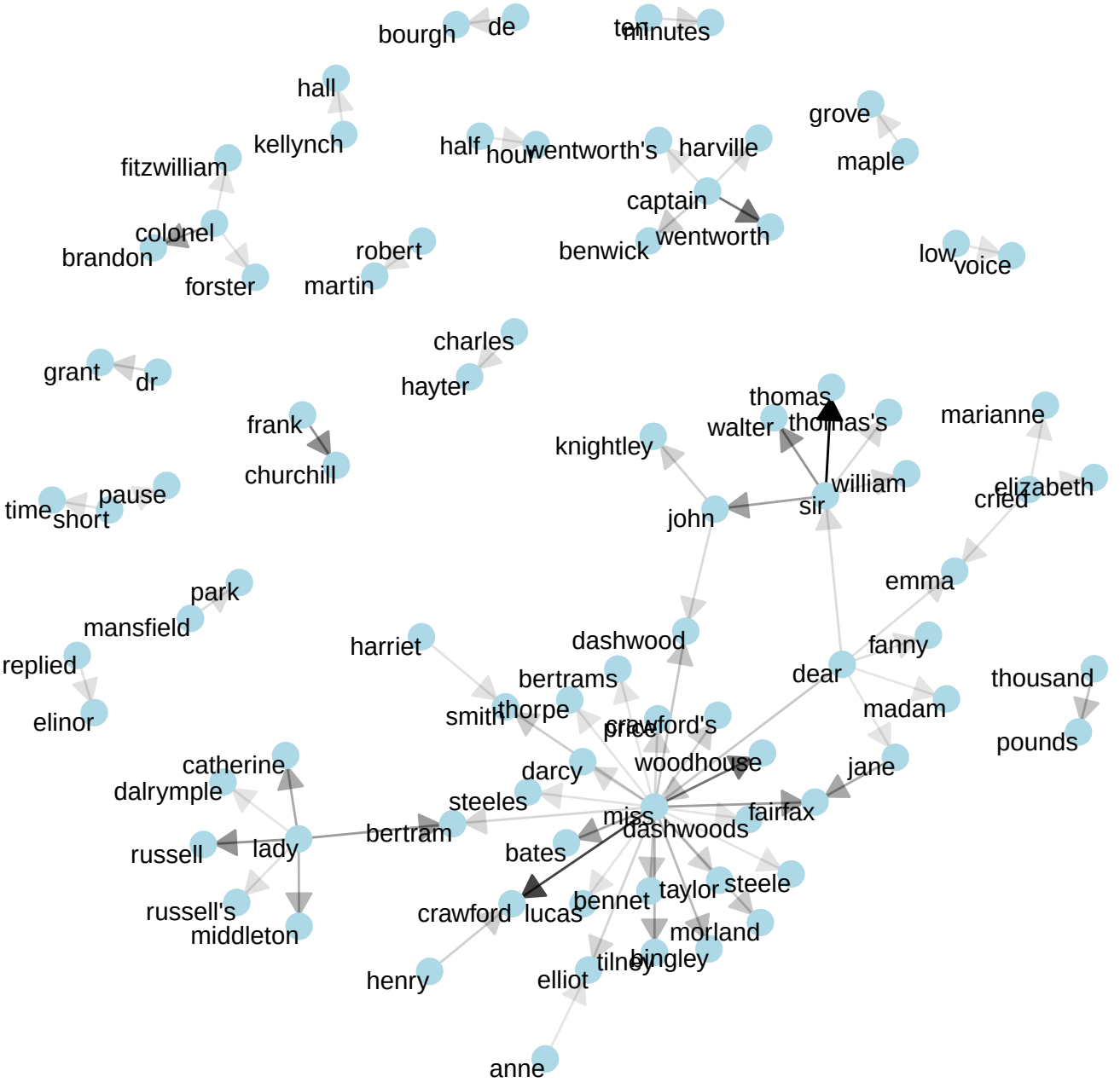

Galilei, Galileo
water
hath
equall
aristotle
board
grave
sidenote
altitude
natation
gravity
swim
prisme
ebony
cone
rampart
0.000
0.002
0.004
0.006
Huygens, Christiaan
refraction
crystal
ray
spheroid
rays
reflexion
movement
refractions
ethereal
wave
refracted
straight
tangent
transparent
spheroids
minutes
ad
0.000
0.001
0.002
0.003
0.004
Tesla, Nikola
bulb
coil
wire
discharge
currents
conducting
frequencies
frequency
electrode
current
wires
impulses
globe
condenser
terminal
button
0.000
0.002
0.004
0.006
Einstein, Albert
relativity
theory
gravitational
ordinates
ordinate
system
continuum
dimensional
transformation
euclidean
embankment
carriage
lorentz
galileian
relative
0.0000
0.0025
0.0050
0.0075
tf−idf

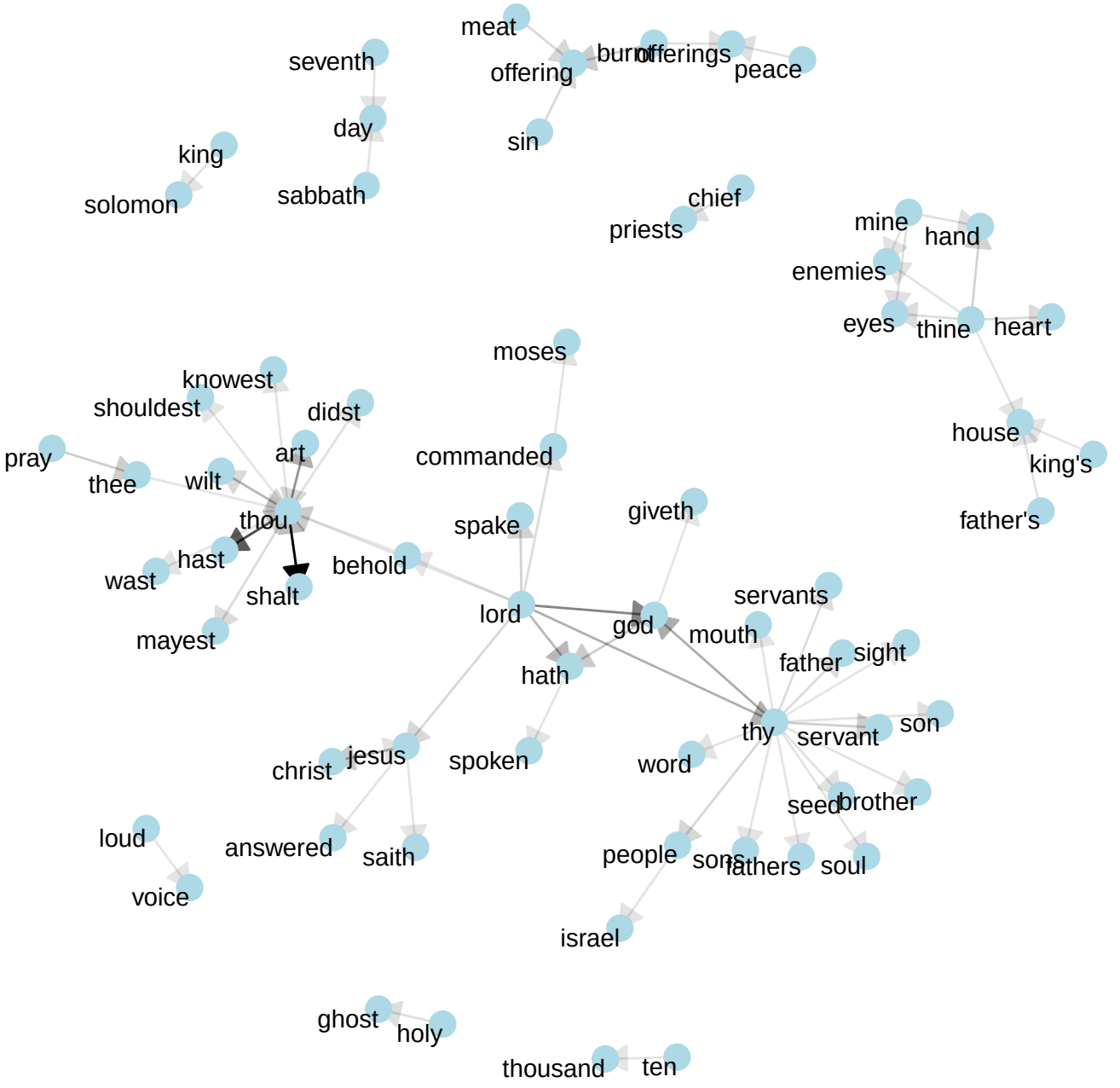

Sense & Sensibility
colonel brandon
sir john
lady middleton
miss dashwood
john dashwood
replied elinor
miss steeles
cried marianne
miss steele
miss dashwoods
0.00
0.01
0.02
Pride & Prejudice
lady catherine
miss bingley
miss bennet
miss darcy
de bourgh
miss lucas
colonel forster
sir william
colonel fitzwilliam
miss de
0.000
0.005
0.010
0.015
0.020
0.025
Mansfield Park
sir thomas
miss crawford
lady bertram
henry crawford
miss crawford's
dr grant
miss price
miss bertram
sir thomas's
mansfield park
0.00
0.02
0.04
Emma
miss woodhouse
frank churchill
miss fairfax
miss bates
jane fairfax
john knightley
miss taylor
dear emma
miss smith
maple grove
0.00
0.01
0.02
0.03
Northanger Abbey
miss tilney
miss morland
miss thorpe
captain tilney
dear catherine
john thorpe
henry tilney
cried catherine
dearest catherine
miss morland's
0.00
0.01
0.02
0.03
Persuasion
captain wentworth
lady russell
sir walter
captain benwick
miss elliot
charles hayter
captain harville
lady russell's
captain wentworth's
kellynch hall
0.00
0.01
0.02
0.03
0.04
0.05
tf−idf of bigram

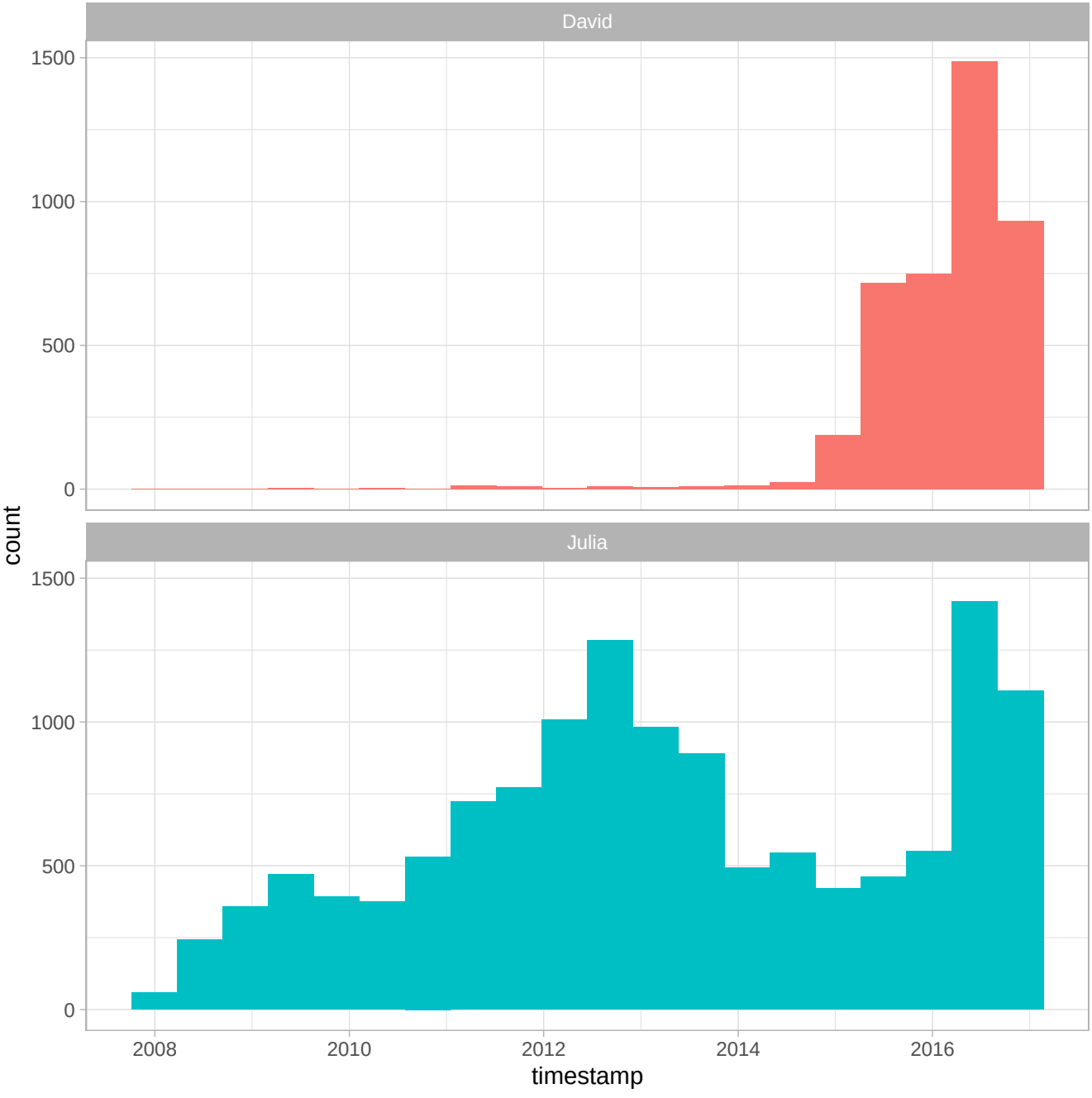

like
help
love
care
want
wish
worth
allow
perfectly
fair
regret
refuse
pretend
leave
despair
deceive
sorry
miss
fail
afraid
Words preceded by "not"
0
50
100
150
200
Sentiment value * number of occurrences

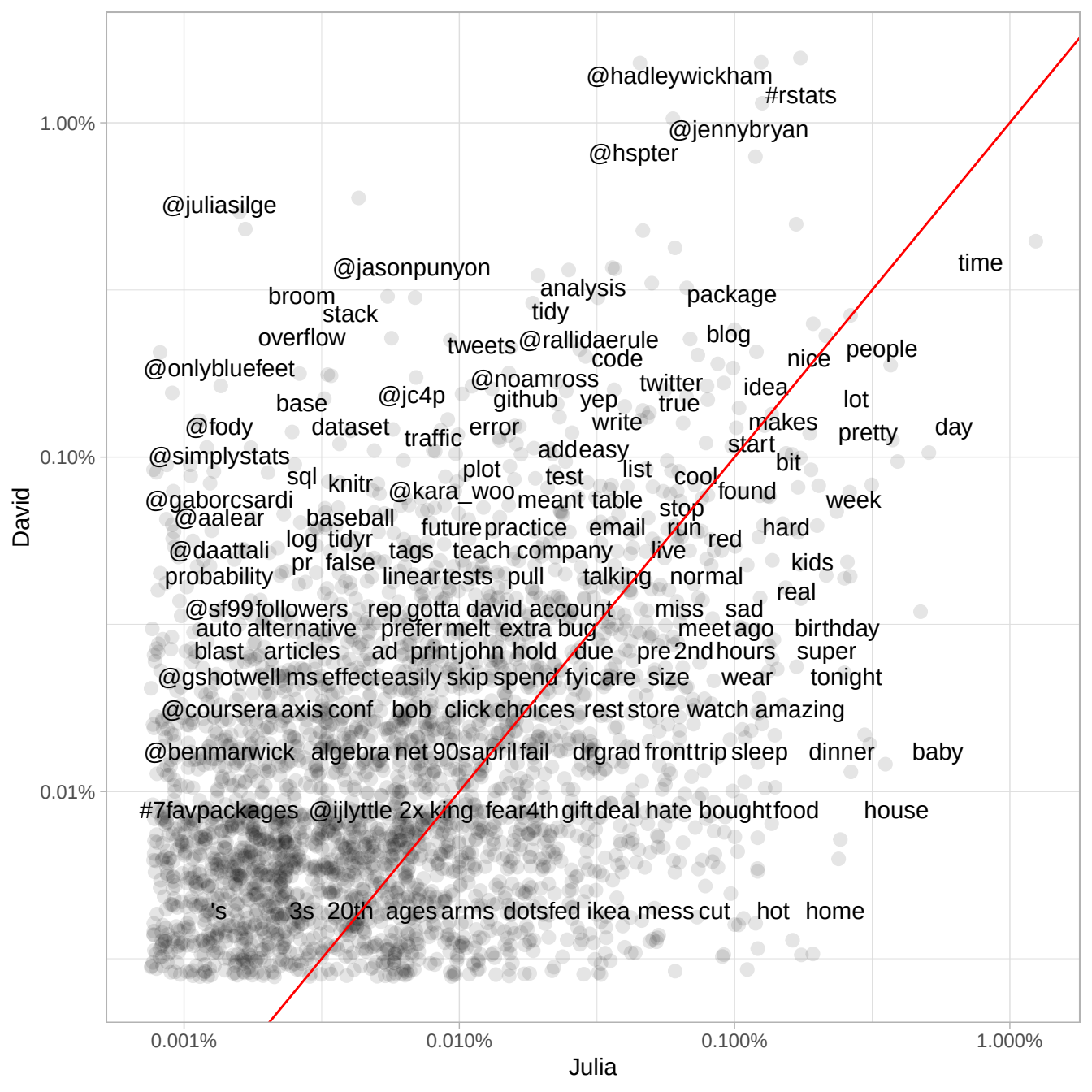

never
loved
want
liked
happy
allow
like
hope
consent
agree
failing
forget
failed
−15
−10
−5
0
5
10
no
great
pleasure
good
hope
chance
better
evil
bad
danger
harm
no
doubt
−100
−50
0
50
not
like
help
love
care
want
wish
worth
allow
perfectly
sorry
fail
afraid
0
50
100
150
200
without
feeling
great
love
hopes
hope
affection
anger
delay
loss
losing
interruption
betraying
−10
0
10
Words preceded by negation term
Sentiment value * # of occurrences

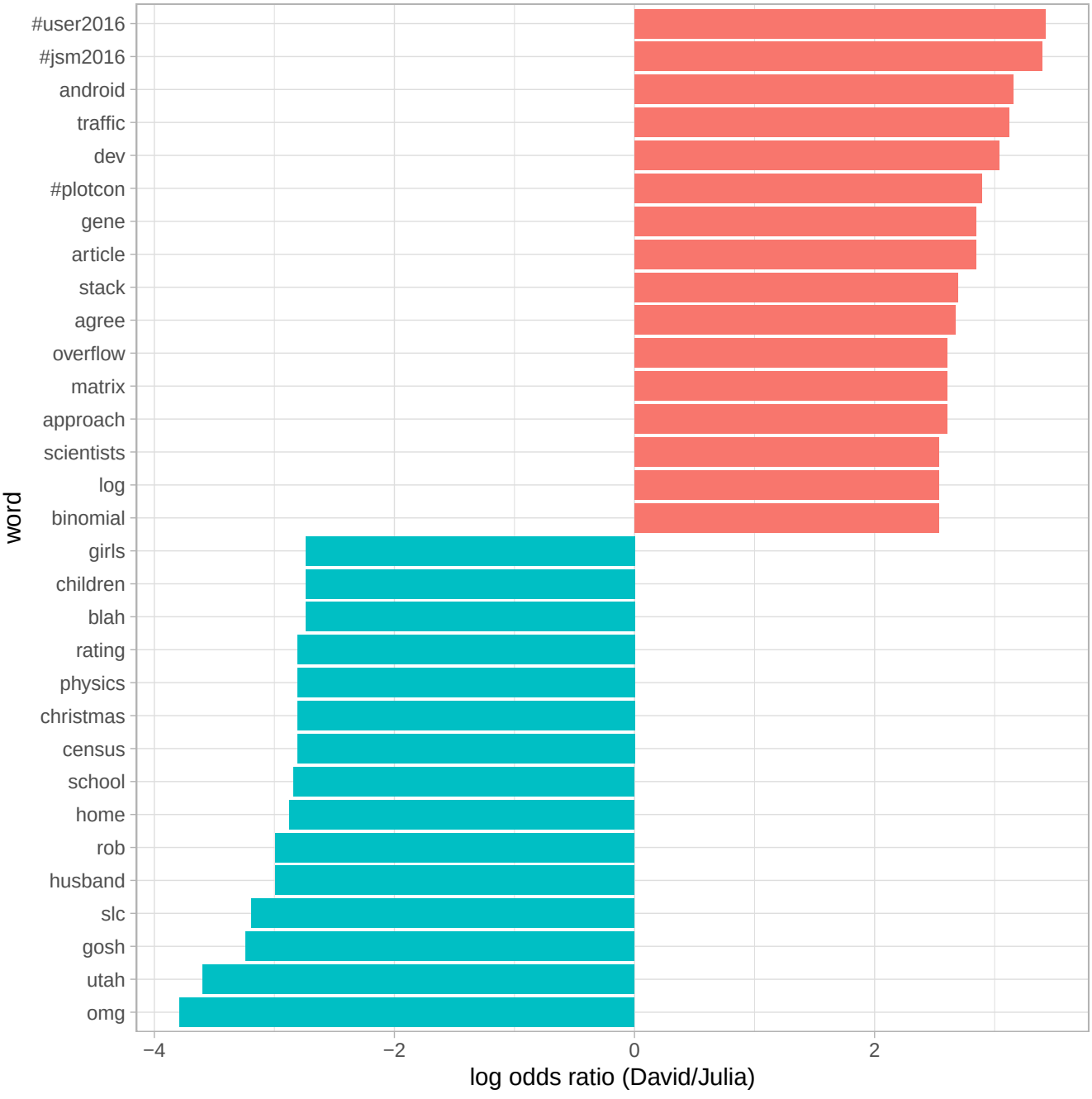

harville
wentworth
captain
benwick
wentworth's
harriet
anne
smith
morland
thorpe
taylor
elliot
catherine
russell's
thousand
bennet
woodhouse
price
churchill
pounds
tilney
steele
steeles
bertram
lady russell
frank
crawford's
miss
bingley
lucas
dashwoods
middleton
dalrymple
crawford
darcy
bertrams
replied
bates
dashwood
henry
elinor
fairfax
rove
john
knightley
jane
dear
bourgh
hour
hall
minutes
ten
madam
fanny
sir thomas
de
park
emma
walter
mansfield
william
thomas's
kellynch
hall
voice
fitzwilliam
cried
low
charles
colonel
elizabeth
marianne
brandon
hayter
forster
robert
martin
time
short
pause
dr
grant

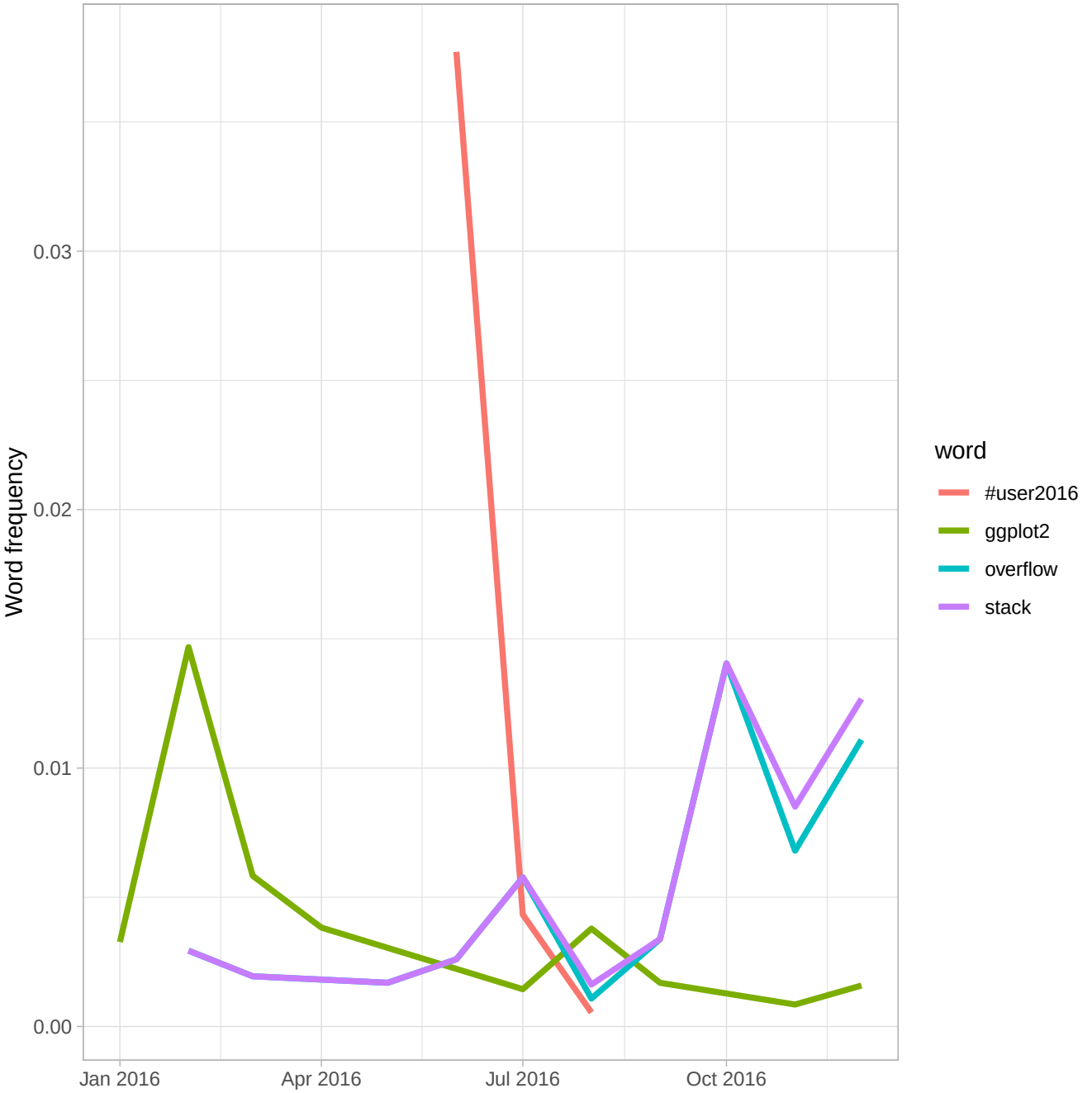

bourgh
de
ten
minutes
hall
grove
kellynch
half
hour
wentworth's
harville
maple
fitzwilliam
captain
colonel
wentworth
brandon
robert
benwick
low
voice
forster
martin
charles
grant
dr
hayter
thomas
marianne
frank
walter
thomas's
knightley
churchill
william
elizabeth
pause
cried
time
short
sir
john
emma
park
mansfield
harriet
dashwood
fanny
replied
bertrams
dear
thousand
elinor
smith
thorpe
madam
price
crawford's
pounds
catherine
darcy
woodhouse
jane
dalrymple
steeles
fairfax
miss
bertram
dashwoods
russell
lady
bates
taylor
steele
russell's
crawford
lucas
bennet
middleton
morland
bingley
tilney
henry
elliot
anne

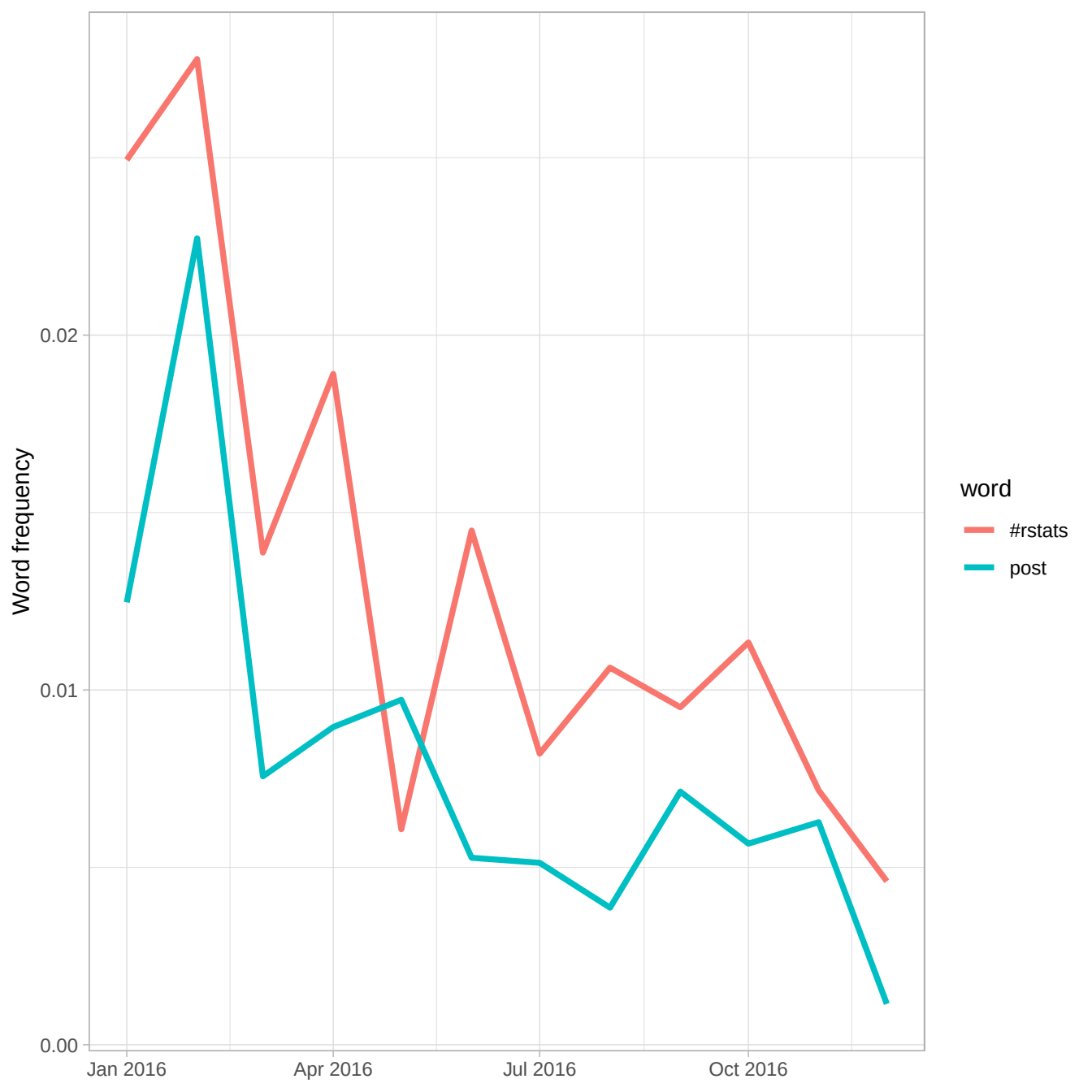

meat
seventh
offering
burnt
offerings
peace
day
sin
king
solomon
sabbath
chief
priests
mine
hand
enemies
eyes
thine
heart
moses
knowest
shouldest
didst
house
pray
art
commanded
king's
thee
wilt
thou
spake
giveth
father's
hast
behold
wast
shalt
servants
lord
god
mouth
mayest
father
sight
hath
thy
son
servant
christ
jesus
spoken
word
seed
brother
loud
answered
saith
people
sons
fathers
soul
voice
israel
ghost
holy
thousand
ten

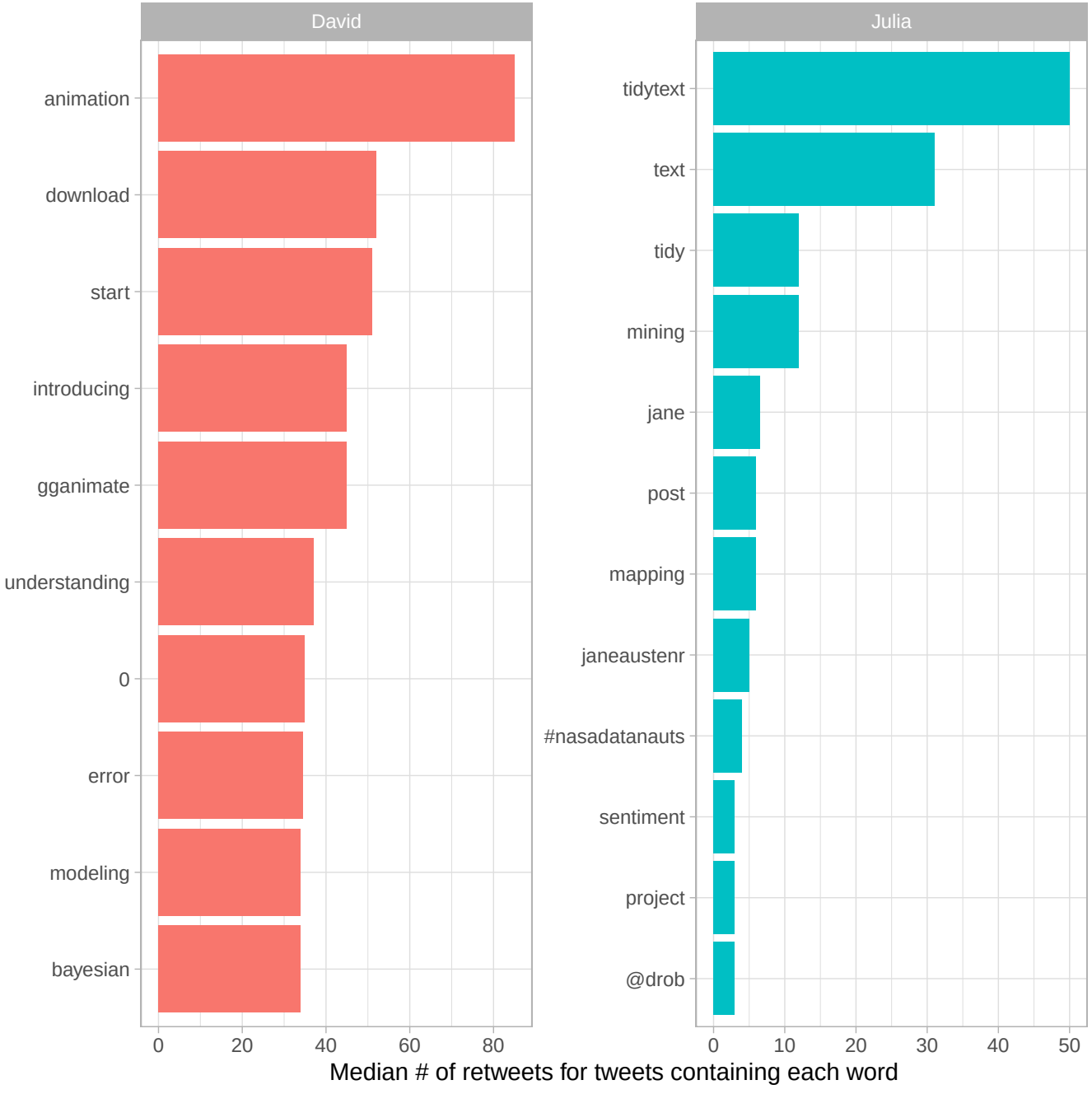

David
Julia
count
1500
1000
500
0
2008
2010
2012
2014
2016
timestamp

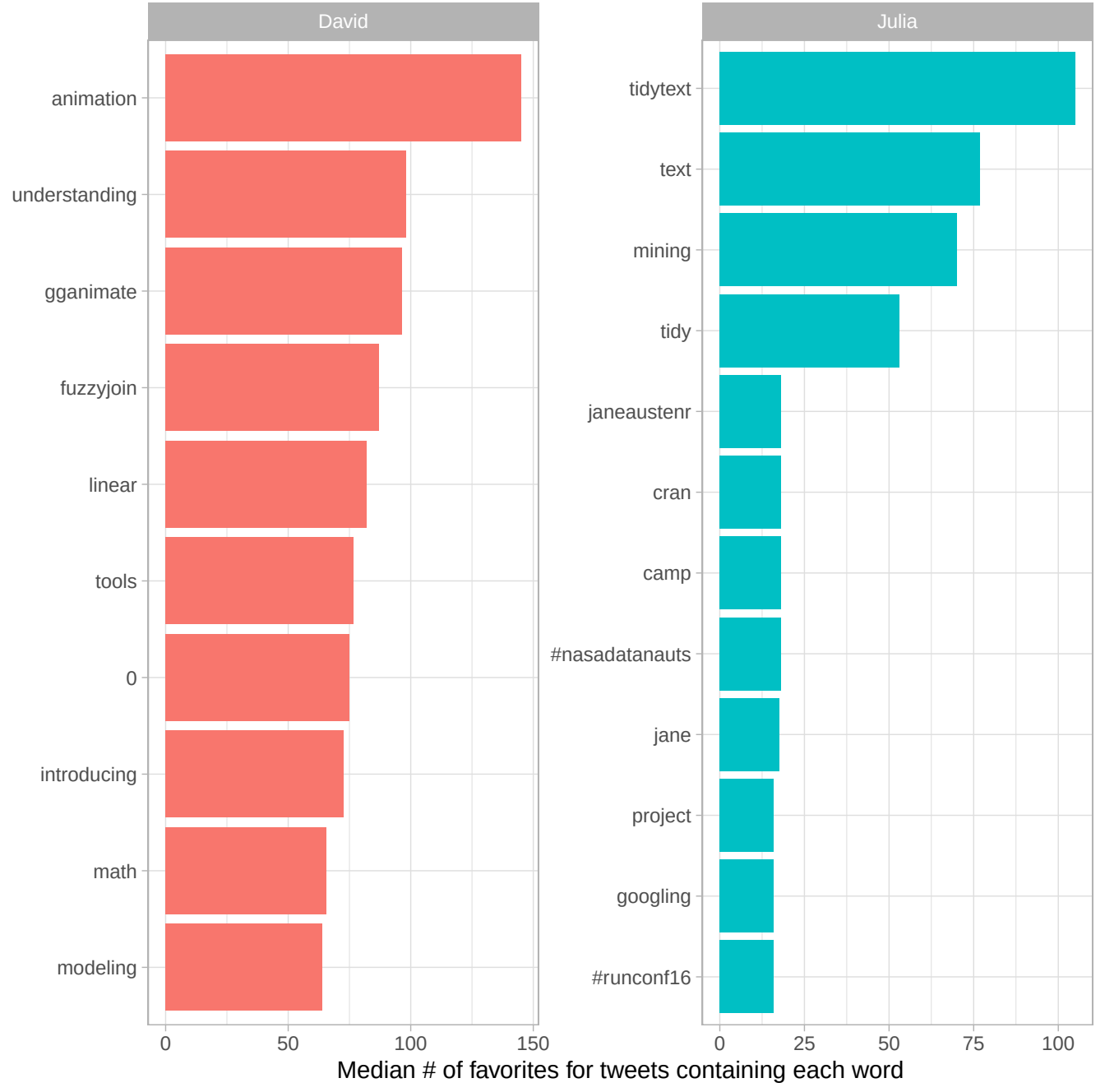

@hadleywickham
#rstats
@jennybryan
@hspter
@juliasilge
@jasonpunyon
time
broom
analysis
package
stack
tidy
overflow
tweets
@rallidaerule
blog
code
nice
people
@onlybluefeet
@noamross
twitter
idea
base
@jc4p
github
yep
true
lot
@fody
dataset
error
write
makes
pretty
day
traffic
start
@simplystats
add
easy
bit
plot
list
sql
knitr
test
cool
@kara_woo
found
@gaborcsardi
meant
table
week
@aalear
baseball
stop
future
practice
email
run
hard
log
tidyr
red
@daattali
tags
teach
company
live
pr
false
kids
probability
linear
tests
pull
talking
normal
real
@sf99
followers
rep
gotta
david
account
miss
sad
auto
alternative
prefer
melt
extra
bug
meet
ago
birthday
blast
articles
ad
print
john
hold
due
pre
2nd
hours
super
@gshotwell
ms
effect
easily
skip
spend
fyi
care
size
wear
tonight
@coursera
axis
conf
bob
click
choices
rest
store
watch
amazing
@benmarwick
algebra
net
90s
april
fail
drgrad
front
trip
sleep
dinner
baby
#7favpackages
@ijlyttle
2x
king
fear
4th
gift
deal
hate
bought
food
house
's
3s
20th
ages
arms
dots
fed
ikea
mess
cut
hot
home
1.00%
0.10%
0.01%
David
0.001%
0.010%
0.100%
1.000%
Julia


#user2016
#jsm2016
android
traffic
dev
#plotcon
gene
article
stack
agree
overflow
matrix
approach
scientists
log
binomial
girls
children
blah
rating
physics
christmas
census
school
home
rob
husband
slc
gosh
utah
omg
word
−4
−2
0
2
log odds ratio (David/Julia)


Word frequency
0.03
0.02
0.01
0.00
Jan 2016
Apr 2016
Jul 2016
Oct 2016
word
#user2016
ggplot2
overflow
stack


Word frequency
0.02
0.01
0.00
Jan 2016
Apr 2016
Jul 2016
Oct 2016
word
#rstats
post


David
animation
download
start
introducing
gganimate
understanding
0
error
modeling
bayesian
0
20
40
60
80
Julia
tidytext
text
tidy
mining
jane
post
mapping
janeaustenr
#nasadatanauts
sentiment
project
@drob
0
10
20
30
40
50
Median # of retweets for tweets containing each word


David
animation
understanding
gganimate
fuzzyjoin
linear
tools
0
introducing
math
modeling
0
50
100
150
Julia
tidytext
text
mining
tidy
janeaustenr
cran
camp
#nasadatanauts
jane
project
googling
#runconf16
0
25
50
75
100
Median # of favorites for tweets containing each word